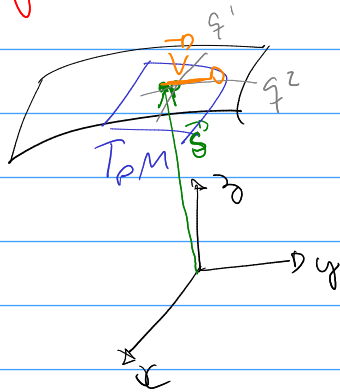


Formalismo Lagrangeano e mov. sobre uma 2-superfície

→ Partícula Livre:

$$L = \frac{1}{2} m v^2$$



$$v^2 = \vec{v} \cdot \vec{v} \quad e \quad \vec{v} = \frac{\partial \vec{S}}{\partial t} \quad d\vec{S} = \vec{e}_1 dq^1 + \vec{e}_2 dq^2$$

Logo: $\vec{v} = \vec{e}_1 \dot{q}^1 + \vec{e}_2 \dot{q}^2 = \vec{e}_i \dot{q}^i$

$$\vec{v} \cdot \vec{v} = (\vec{e}_i \dot{q}^i) (\vec{e}_j \dot{q}^j) = (\underbrace{\vec{e}_i \cdot \vec{e}_j}_{g_{ij}}) \dot{q}^i \dot{q}^j \Rightarrow \vec{v} \cdot \vec{v} = g_{ij} \dot{q}^i \dot{q}^j$$

Equações de movimento:

$$\underbrace{\frac{\partial L}{\partial q^k}}_{1^\circ \text{ termo}} - \underbrace{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^k} \right)}_{2^\circ \text{ termo}} = 0$$

P/ $L = \frac{1}{2} m g_{ij} \dot{q}^i \dot{q}^j$

1º termo:

$$\frac{\partial L}{\partial q^k} = \frac{\partial}{\partial q^k} \left(\frac{1}{2} m g_{ij} \dot{q}^i \dot{q}^j \right) = \frac{m}{2} \frac{\partial g_{ij}}{\partial q^k} \dot{q}^i \dot{q}^j$$

2º termo:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^k} \right) = \frac{d}{dt} \left[\frac{\partial}{\partial \dot{q}^k} \left(\frac{1}{2} m g_{ij} \dot{q}^i \dot{q}^j \right) \right] = \frac{d}{dt} \left\{ \frac{m}{2} g_{ij} \left(\underbrace{\frac{\partial \dot{q}^i}{\partial \dot{q}^k}}_{=\delta^i_k} \dot{q}^j + \dot{q}^i \frac{\partial \dot{q}^j}{\partial \dot{q}^k} \right) \right\}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^k} \right) = \frac{d}{dt} \left[\frac{m}{2} g_{ij} (\dot{q}^i \dot{q}^j + \dot{q}^j \dot{q}^i) \right]$$

$$= \frac{d}{dt} \left[\frac{m}{2} (g_{ki} \dot{q}^i + g_{ik} \dot{q}^i) \right] = \frac{d}{dt} \left[\frac{m}{2} (g_{ki} \dot{q}^i + g_{ki} \dot{q}^i) \right]$$

inverte

$j \rightarrow i$
(indice mudos)

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^k} \right) = \frac{d}{dt} \left[\frac{m}{2} (g_{ki} \dot{q}^i + g_{ki} \dot{q}^i) \right] = \frac{d}{dt} \left[\frac{m}{2} \cancel{2} g_{ki} \dot{q}^i \right]$$

$$= \frac{d}{dt} (m g_{ki} \dot{q}^i) = m \left(\frac{\partial g_{ki}}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial t} + g_{ki} \ddot{q}^i \right)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^k} \right) = m (\partial_j g_{ki} \dot{q}^j \dot{q}^i + g_{ki} \ddot{q}^i)$$

Juntando tudo:

$$\cancel{\frac{m}{2} \frac{\partial g_{ij}}{\partial \dot{q}^k} \dot{q}^i \dot{q}^j} - \cancel{m (\partial_j g_{ki} \dot{q}^j \dot{q}^i + g_{ki} \ddot{q}^i)} = 0$$

$$\frac{1}{2} \partial_k g_{ij} \dot{q}^i \dot{q}^j - (\partial_j g_{ki} \dot{q}^j \dot{q}^i + g_{ki} \ddot{q}^i) = 0$$

$\rightarrow \frac{1}{2} (\partial_j g_{ki} + \partial_i g_{jk}) \dot{q}^j \dot{q}^i$

$$\frac{1}{2} \partial_k g_{ij} \dot{q}^i \dot{q}^j - \frac{1}{2} (\partial_j g_{ki} + \partial_i g_{jk}) \dot{q}^j \dot{q}^i + g_{ki} \ddot{q}^i = 0$$

$$\frac{1}{2} \dot{q}^i \dot{q}^j (\partial_k g_{ij} - \partial_j g_{ki} + \partial_i g_{jk}) + g_{ki} \ddot{q}^i = 0$$

$\rightarrow i, j = 1, 2$
(superfície)

Defino $[g^{ij}]^{-1} = [g_{ij}]$ t.q. $g^{ik} g_{kj} = \delta^k_j$

$$g^{km} \frac{1}{2} \dot{q}^i \dot{q}^j (\partial_k g_{ij} - \partial_j g_{ki} + \partial_i g_{jk}) + \underbrace{g_{ki} \ddot{q}^i g^{km}}_{= \delta^m_i \ddot{q}^i = \ddot{q}^m} = 0$$

$$\ddot{q}^m + \frac{1}{2} g^{km} (\partial_k g_{ij} - \partial_j g_{ki} + \partial_i g_{jk}) \dot{q}^i \dot{q}^j = 0$$

Simbolo de Christoffel

$$\Gamma^m_{ij}$$

$$\boxed{\ddot{q}^m + \Gamma^m_{ij} \dot{q}^i \dot{q}^j = 0}$$